

A submarine cable, also known as an undersea cable or undersea communication cable, is a specialized type of cable designed for the purpose of transmitting data, telecommunications signals, and sometimes electrical power beneath the ocean or sea. These cables are used to establish high-speed, long-distance communication links between countries and continents.

- **Undersea Connectivity:** Submarine cables are specialized cables laid on the ocean floor to connect different parts of the world, enabling global communication and data exchange.
- **Internet Backbone:** They serve as the backbone of the internet, carrying the vast majority of international internet traffic, including data, voice, and video transmissions.
- **Design and Durability:** Submarine cables are designed to withstand harsh ocean conditions, with multiple protective layers and materials to ensure long-term reliability.
- **Global Collaboration:** The installation and maintenance of submarine cables often involve cooperation between multiple countries due to their international nature.
- **High Capacity:** Submarine cables offer high bandwidth and low latency, supporting the high-speed transmission of data over long distances.
- **Critical Infrastructure:** They are critical infrastructure for global communication, trade, and information sharing, underpinning the functioning of the modern connected world.